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D INFK

Cloud Computing Architecture

Google Cloud Platform (GCP) and Project Overview

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M.Sc. Students

Slides adapted from Foteini Strati

Agenda

1. Project Logistics
2. Intro to Cloud Providers and GCP
3. Cluster setup: Kops
4. Running applications in a cluster: Kubernetes
5. OpenEvolve
6. How to get correct measurements

Project Overview

Goal: Schedule applications of different requirements (latency-sensitive, batch applications) in the cloud efficiently.

- **Part 1:** Measure the effect of resource **interference** (using **iBench**) in **memcached** application
- **Part 2:** Measure the effect of resource **interference** (using **iBench**) and **scalability** of **PARSEC** benchmarks.
- **Part 3:** Use insights from Part 1 & 2 to schedule memcached and PARSEC in **a cluster of heterogeneous VMs (under static memcached load)**.
- **Part 4:** Schedule memcached and PARSEC together (collocated) on **a single machine (under dynamic memcached load)**.

Project Logistics

- Deadline to submit group preferences for the project: **Friday, March 6th, 2026**.
- Groups will be assigned on **Tuesday, March 10th, 2026**. You may then start working on the project.
- Project Part 1 and 2 are **not graded**
- Project deadline for Part 3 and 4: **Saturday, May 16th, 2026, end of day**

- **You MUST signup on Moodle for the project (otherwise you cannot do the project):**
  - Sign up for a **specific** group

    or
  - Sign up for **general** group (and we will place you in a group)

Project Logistics

- Use of AI (LLMs) not prohibited, but
  - It must adhere to ETH regulations
  - This means you take responsibility for the content you submit
  - You must disclose any use of GenAI and other AI tools in your work
  - Avoid sharing copyrighted, or confidential information with commercial AI platforms
  - Failure to comply with these guidelines may result in disciplinary action

- However, it might still be helpful as a tutor or help you write shell scripts (use **with caution**)

- **This year**: Parts 3 and 4 will involve using LLMs to attempt to improve upon hand-crafted policy using the OpenEvolve framework. Instructions on how to set this up can be found in Part 3 of the Project Instructions

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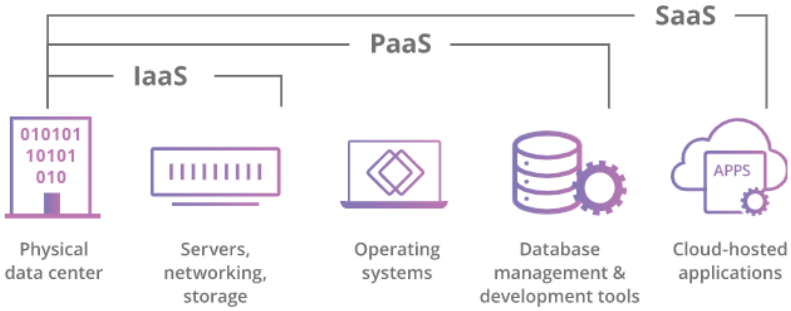
Cloud Providers and GCP

The Major Cloud Providers

- Many public cloud providers, such as:
  - Google Cloud Platform (GCP)
  - Amazon Web Services (AWS)
  - Microsoft Azure
- They provide services such as:
  - **Function as a Service (FaaS):**
    - serverless computing
  - **Software as a Service (SaaS):**
    - cloud gaming, e-mail, …
  - **Platform as a Service (PaaS):**
    - app deployment, querying, …
  - **Infrastructure as a Service (IaaS):**
    - VMs, storage, …



Google Cloud Platform (GCP)

- Offers a set of cloud computing services, such as:
  - Compute Engine
  - Cloud Storage/Filestore
  - BigQuery/DataFlow
  - Cloud Functions
  - AI Platform

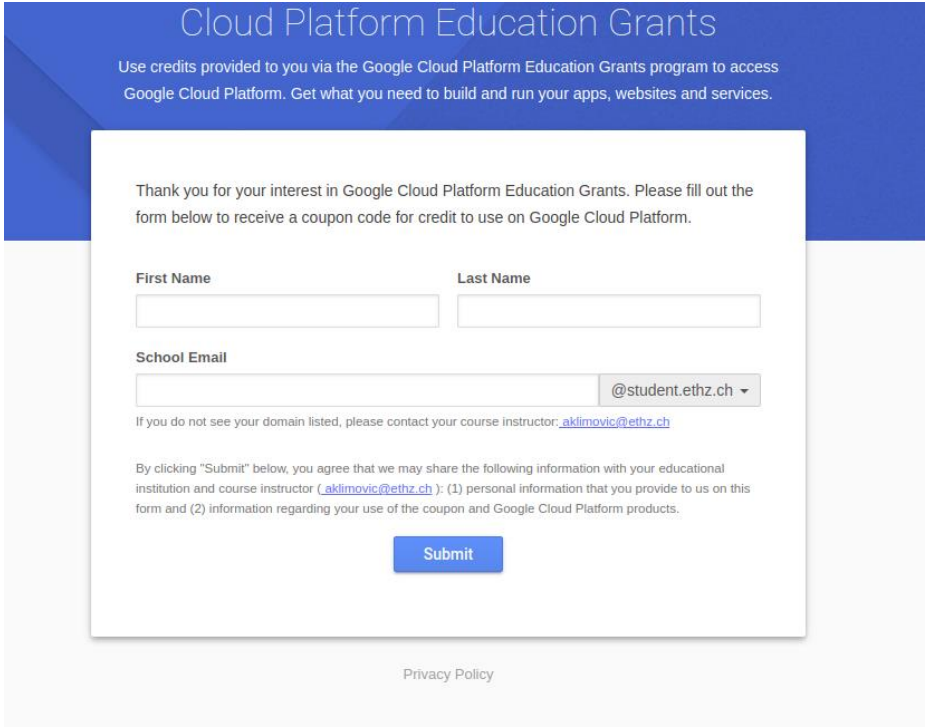
- Same infrastructure that Google uses internally for its products (Gmail, Google Drive, etc.)

GCP - Agenda

- Create a Project in GCP
- Configure GCP CLI
- Credits, Billing & Budget
- Create VMs
- Inspect/Access VMs
- Terminate VMs

GCP - Create Project (Get Credits)

- You must have a Google Cloud account under your **ETH email address**
- **One group member** redeems the cloud credits (instructions in the project handout)
- **You should NOT redeem credits until the official start date of the project!**



GCP - Create Project

- Login and create the project via gcloud

```
$ gcloud init

[...]
You are logged in as: [ethzid@ethz.ch].

Pick cloud project to use:
 [1] …
 …
 [6] Enter a project ID
 [7] Create a new project
Please enter numeric choice or text value (must exactly match list item):  7

Enter a Project ID. Note that a Project ID CANNOT be changed later.
Project IDs must be 6-30 characters (lowercase ASCII, digits, or
hyphens) in length and start with a lowercase letter. cca-eth-2026-group-999
…
Your current project has been set to: [cca-eth-2026-group-999].
```

GCP - Create Project

- Link the project with the **Billing Account for Education**

Set the billing account for project “cca-eth-2023-group-123”

Billing account *

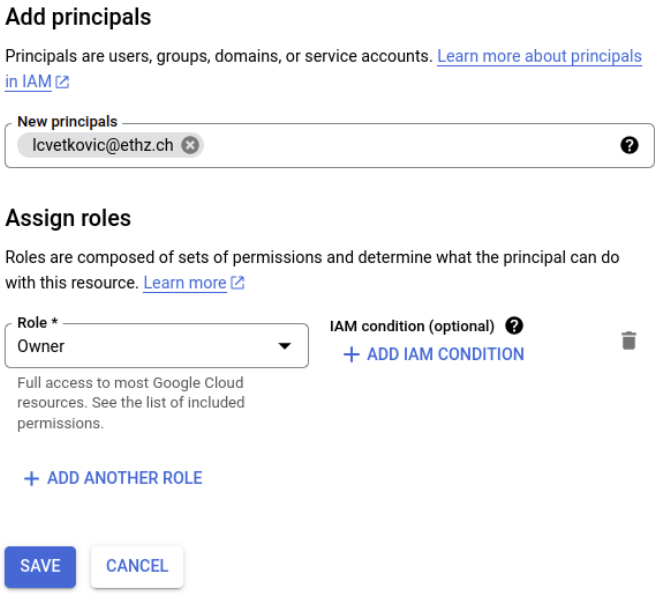
Billing Account for Education

Any charges for this project will be billed to the account you select here.

CANCEL SET ACCOUNT

GCP - Create Project

- Add your teammates to the project



GCP - Create Project

- View available zones (usually have all VMs in the same availability zone)

```
$ gcloud compute zones list

API [compute.googleapis.com] not enabled on project [...]. Would you like to enable and retry (this will take a
few minutes)? (y/N)?  y

Enabling service [compute.googleapis.com] on project [...]...

NAME              REGION          STATUS
us-east1-b         us-east1        UP
us-east1-c         us-east1        UP
us-east1-d         us-east1        UP
us-east4-c         us-east4        UP
us-east4-b         us-east4        UP
us-east4-a         us-east4        UP
us-central1-c      us-central1     UP
....
```

Shown the first time

GCP - Configure CLI

Get available/active accounts:

```
$ gcloud auth list

Credentialed Accounts
ACTIVE  ACCOUNT
*       username@ethz.ch

To set the active account, run:
  $ gcloud config set account `ACCOUNT`
```

Set/get current

```
$ gcloud config set project myproject

Updated property [core/project].


$ gcloud config get-value project

myproject
```

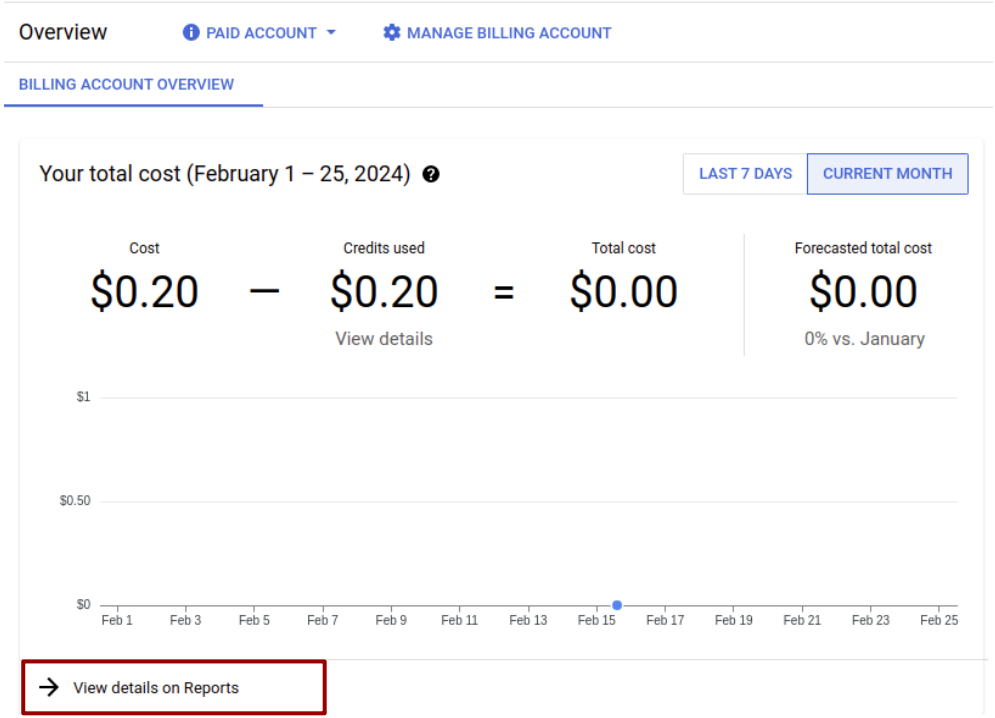
Set account

```
$ gcloud auth application-default login

Your browser has been opened to visit:
....
```

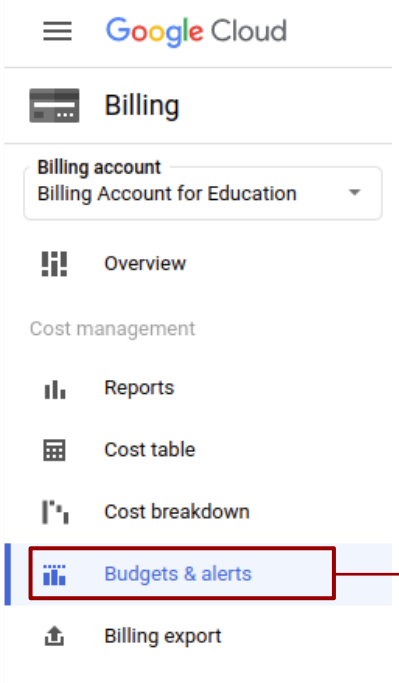
GCP - Credits, Billing & Budget

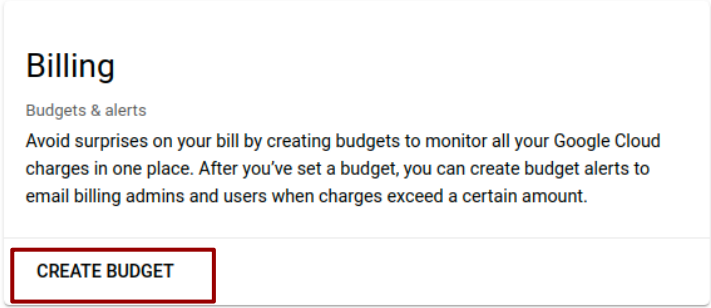
- View remaining credits/cost overview



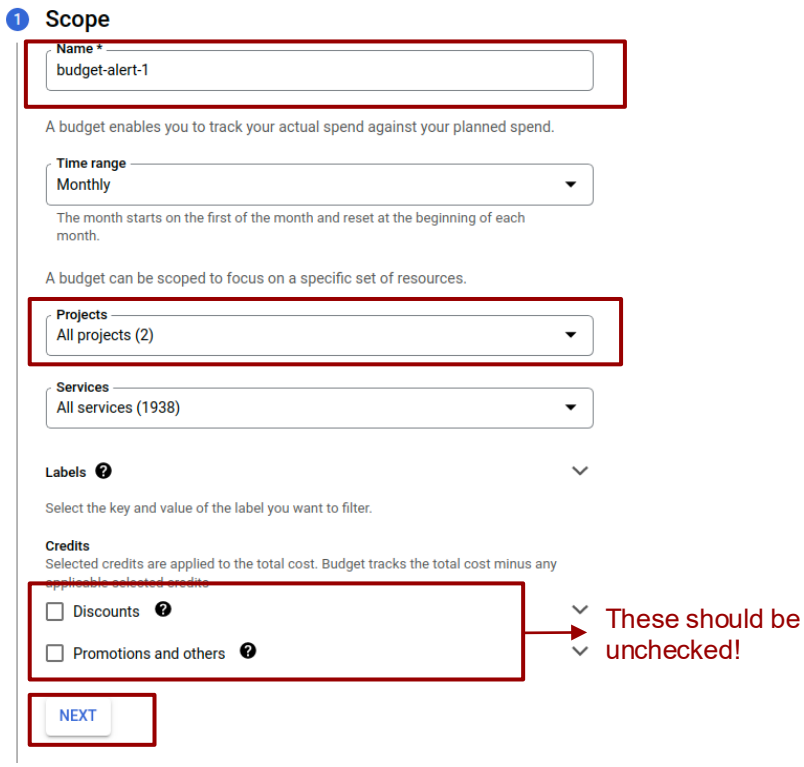
GCP - Credits, Billing & Budget

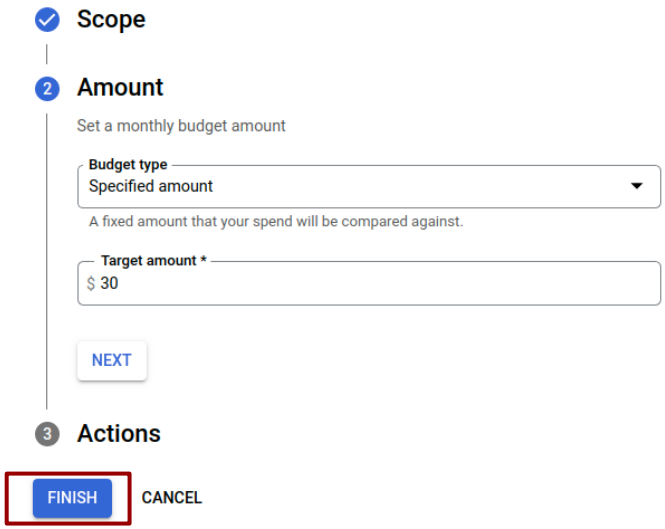
- Budget alerts





GCP - Credits, Billing & Budget





GCP - Create VM - CLI

- Get a list of available images

```
$ gcloud compute images list

NAME                                  PROJECT        FAMILY            DEPRECATED   STATUS

centos-7-v20220126                    centos-cloud   centos-7                       READY
centos-stream-8-v20220128             centos-cloud   centos-stream-8                READY
cos-85-13310-1416-5                   cos-cloud      cos-85-lts                     READY
cos-89-16108-604-11                   cos-cloud      cos-89-lts                     READY
cos-93-16623-102-12                   cos-cloud      cos-93-lts                     READY
cos-beta-93-16623-39-6                cos-cloud      cos-beta                       READY
debian-10-buster-v20220118            debian-cloud   debian-10                      READY
debian-11-bullseye-v20220120          debian-cloud   debian-11                      READY
debian-9-stretch-v20220118            debian-cloud   debian-9                       READY
fedora-cloud-base-gcp-33-1-2-x86-64   fedora-cloud   fedora-cloud-33                READY
fedora-cloud-base-gcp-34-1-2-x86-64   fedora-cloud   fedora-cloud-34                READY
fedora-cloud-base-gcp-35-1-2-x86-64   fedora-cloud   fedora-cloud-35                READY
….
```

GCP - Create VM - CLI

- Create the VM

```
$ gcloud compute instances create vm-gcloud --image=ubuntu-2204-jammy-v20240208 --image-project=ubuntu-os-cloud --machine-type=n2d-highcpu-2 --project=cca-eth-2024-group-999 --zone=europe-west3-a

Created [https://www.googleapis.com/compute/v1/projects/cca-eth-2024-group-999/zones/europe-west3-a/instances/vm-gcloud].


NAME        ZONE            MACHINE_TYPE    PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP     STATUS
vm-gcloud   europe-west3-a  n2d-highcpu-2                10.156.0.2   35.246.199.121  RUNNING
```

GCP - Create VM - Console

Google Cloud | cca-eth-2024-group-999

Cloud overview
Products & solutions

PINNED PRODUCTS
Compute Engine

VIRTUAL MACHINES
VM instances
Instance templates
Sole-tenant nodes
Machine images
TPUs

VM instances → CREATE INSTANCE | IMPORT VM | REFRESH

Name *
vm-console

MANAGE TAGS AND LABELS

Region *
europe-west3 (Frankfurt)
Region is permanent

Zone *
europe-west3-a
Zone is permanent

Machine configuration

NEW: General-purpose machine series in Preview
Try the new N4 series, ideal for workloads that prioritize flexibility and cost-optimization
SIGN UP

General purpose | Compute optimized | Memory optimized | Storage optimized NEW | GPUs

Machine types for common workloads, optimized for cost and flexibility

Series	Description	vCPUs	Memory	Platform
C3	Consistently high performance	4 - 176	8 - 1,408 GB	Intel Sapphire Rapids
C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Based on availability
N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade and Ice Lake
N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD EPYC
T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra Arm
T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMD EPYC Milan
N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB	Intel Skylake

Machine type
Choose a machine type with preset amounts of vCPUs and memory that suit most workloads.
Or, you can create a custom machine for your workload's particular needs. Learn more

PRESET | CUSTOM

n2d-highcpu-2 (2 vCPU, 1 core, 2 GB memory)

vCPU	Memory
2 (1 core)	2 GB

Boot disk

Name	vm-console
Type	New balanced persistent disk
Size	10 GB
License type	Free
Image	Ubuntu 22.04 LTS

Select disk image

CHANGE

CREATE | CANCEL | EQUIVALENT CODE

GCP - Inspect VM

- From CLI

```
$ gcloud compute instances list

NAME        ZONE            MACHINE_TYPE  PREEMPTIBLE  INTERNAL_IP   EXTERNAL_IP      STATUS
vm-console  europe-west3-a  n2d-highcpu-2              10.156.0.3    34.159.195.245   RUNNING
vm-gcloud   europe-west3-a  n2d-highcpu-2              10.156.0.2    35.246.199.121   RUNNING
```

- From Console

INSTANCES OBSERVABILITY INSTANCE SCHEDULES

VM instances

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
✓	vm-console	europe-west3-a			10.156.0.3 (nic0)	34.159.195.245 (nic0)	SSH
✓	vm-gcloud	europe-west3-a			10.156.0.2 (nic0)	35.246.199.121 (nic0)	SSH

GCP - Connect to VM

- Connect via SSH

```
$ gcloud compute ssh vm-console --project cca-eth-2024-group-999 --zone europe-west3-a

[...]

fot@vm-console:~$
```

GCP - Terminate VM

- From CLI

```
$ gcloud compute instances delete vm-console --project cca-eth-2024-group-999 --zone europe-west3-a

The following instances will be deleted. Any attached disks configured to be auto-deleted will be deleted unless they are
attached to any other instances or the `--keep-disks` flag is given and specifies
them for keeping. Deleting a disk is irreversible and any data on the disk will be lost.
 - [vm-console] in [europe-west3-a]

Do you want to continue (Y/n)?  Y

Deleted [https://www.googleapis.com/compute/v1/projects/cca-eth-2024-group-999/zones/europe-west3-a/instances/vm-console].
```

GCP - Terminate VM

- From Console

VM instances | CREATE INSTANCE | IMPORT VM | REFRESH

INSTANCES | OBSERVABILITY | INSTANCE SCHEDULES

1 instance selected | START / RESUME | STOP | SUSPEND | RESET | DELETE | LABELS | PERMISSIONS

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
	vm-gcloud	europe-west3-a			10.156.0.2 (nic0)	35.246.199.121 (nic0)	SSH

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Setting up a cluster with Kops

Kops

- Tool to create, maintain, update and delete Kubernetes-enabled clusters [1]
- Users define a *cluster specification* (e.g. a yaml file)
  - We will describe how our GCP VMs should look like (number of machines, machine type, etc.)
- Supports multiple cloud providers such as AWS, GCP

[1] Kops guide

Kops yaml file

```
apiVersion: kops.k8s.io/v1alpha2
kind: Cluster
metadata:
  creationTimestamp: null
  name: part1.k8s.local
spec:
  …
  configBase: gs://cca-eth-2024-group-XXX-ethzid/part1.k8s.local
  …
  project:  cca-eth-2024-group-XXX
  …
```

Change these!

```
apiVersion: kops.k8s.io/v1alpha2
kind: InstanceGroup
metadata:
  creationTimestamp: null
  labels:
    kops.k8s.io/cluster: part1.k8s.local
  name: memcache-server
spec:
  image: ubuntu-os-cloud/ubuntu-2204-jammy-v20240208
  machineType: n2d-highcpu-2
  maxSize: 1
  minSize: 1
  nodeLabels:
    cloud.google.com/metadata-proxy-ready: "true"
    kops.k8s.io/instancegroup: nodes-europe-west3-a
    cca-project-nodetype: "memcached"
…
```

Create a cluster with Kops

```
$ gsutil mb gs://cca-eth-2024-group-XXX-ethzid/
$ export KOPS_STATE_STORE=gs://cca-eth-2024-group-XXX-ethzid/
```

For the cluster configuration

```
$ cd ~/.ssh & ssh-keygen -t rsa -b 4096 -f cloud-computing
```

Generate login keys

```
$ PROJECT=`gcloud config get-value project`
```

```
$ kops create -f part1.yaml
```

Set configuration

```
$ kops create secret --name part1.k8s.local sshpublickey admin -i ~/.ssh/cloud-computing.pub
```

Add login keys

```
$ kops update cluster --name part1.k8s.local --yes --admin
```

Deploy the cluster

```
$ kops validate cluster --wait 10m
```

Wait…

Inspect the cluster

$ kubectl get nodes -o wide

NAME	STATUS	ROLES	AGE	VERSION	INTERNAL-IP	EXTERNAL-IP	OS-IMAGE	KERNEL-VERSION	CONTAINER-RUNTIME
client-agent-q6j6	Ready	node	2m12	v1.28.6	10.0.16.3	35.246.168.138	Ubuntu 22.04.3 LTS	6.5.0-1013-gcp	containerd://1.7.13
client-measure-37f6	Ready	node	2m28	v1.28.6	10.0.16.4	34.159.157.249	Ubuntu 22.04.3 LTS	6.5.0-1013-gcp	containerd://1.7.13
master-europe-west3-a-r5nf	Ready	control-plane	3m43	v1.28.6	10.0.16.5	34.141.90.165	Ubuntu 22.04.3 LTS	6.5.0-1013-gcp	containerd://1.7.13
memcache-server-24bd	Ready	node	2m31	v1.28.6	10.0.16.2	34.89.190.131	Ubuntu 22.04.3 LTS	6.5.0-1013-gcp	containerd://1.7.13

VM instances

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
✓	client-agent-q6j6	europe-west3-a		a-client-agent-part1-k8s-local	10.0.16.3 (nic0)	35.246.168.138 (nic0)	SSH
✓	client-measure-37f6	europe-west3-a		a-client-measure-part1-k8s-local	10.0.16.4 (nic0)	34.159.157.249 (nic0)	SSH
✓	master-europe-west3-a-r5nf	europe-west3-a		a-master-europe-west3-a-part1-k8s-local, api-part1-k8s-local	10.0.16.5 (nic0)	34.141.90.165 (nic0)	SSH
✓	memcache-server-24bd	europe-west3-a		a-memcache-server-part1-k8s-local	10.0.16.2 (nic0)	34.89.190.131 (nic0)	SSH

Instance Templates

Instance templates

CREATE INSTANCE TEMPLATE | REFRESH | CREATE VM | CREATE INSTANCE GROUP | COPY | DELETE | LEARN | SHOW INFO PANEL

Instance templates are saved VM configurations used to create identical VMs, either individually or as part of managed instance groups. Learn more

Filter Filter instance templates

Name	Machine type	Image	Disk type	Location	Placement policy	In use by	Creation time	Actions
client-agent-part1-k8s-local-1708893251	e2-standard-16	ubuntu-2204-jammy-v20240208	Standard persistent disk	global	No policy	a-client-agent-part1-k8s-local	Feb 25, 2024, 9:34:13 PM UTC+01:00	
client-measure-part1-k8s-local-1708893251	e2-standard-2	ubuntu-2204-jammy-v20240208	Standard persistent disk	global	No policy	a-client-measure-part1-...	Feb 25, 2024, 9:34:12 PM UTC+01:00	
master-europe-west3-a-par-eeo8bm-1708893251	e2-standard-2	ubuntu-2204-jammy-v20240208	Standard persistent disk	global	No policy	a-master-europe-west3...	Feb 25, 2024, 9:34:13 PM UTC+01:00	
memcache-server-part1-k8s-local-1708893251	n2d-highcpu-2	ubuntu-2204-jammy-v20240208	Standard persistent disk	global	No policy	a-memcache-server-pa...	Feb 25, 2024, 9:34:14 PM UTC+01:00	

Load Balancers

Load balancing

+ CREATE LOAD BALANCER REFRESH DELETE

LOAD BALANCERS BACKENDS FRONTENDS

Filter Enter property name or value

Name	Load balancer type	Access type	Protocols	Region	Backends
api-part1-k8s-local	Network (Passthrough target-pool)	External	TCP	europe-west3	1 target pool (1 instance)

To view or delete load balancing resources like forwarding rules and target proxies, go to the load balancing components view.

Delete the cluster

```
$ kops delete cluster part1.k8s.local --yes

[...]
Deleted kubectl config for part1.k8s.local
Deleted cluster: "part1.k8s.local"
```

It is very important to delete your cluster,

otherwise you may spend your credits quite quickly!

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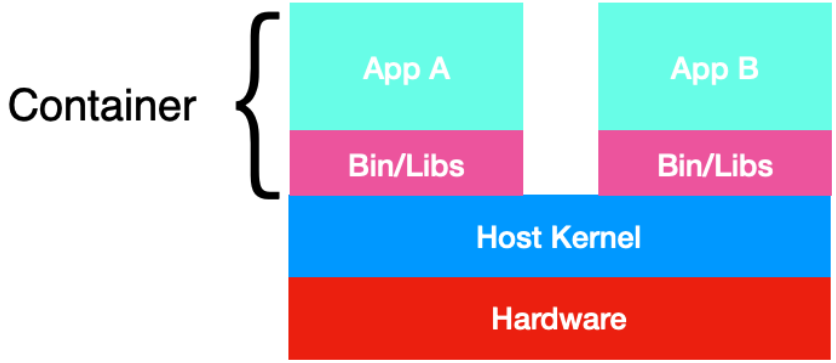
Running applications with Kubernetes

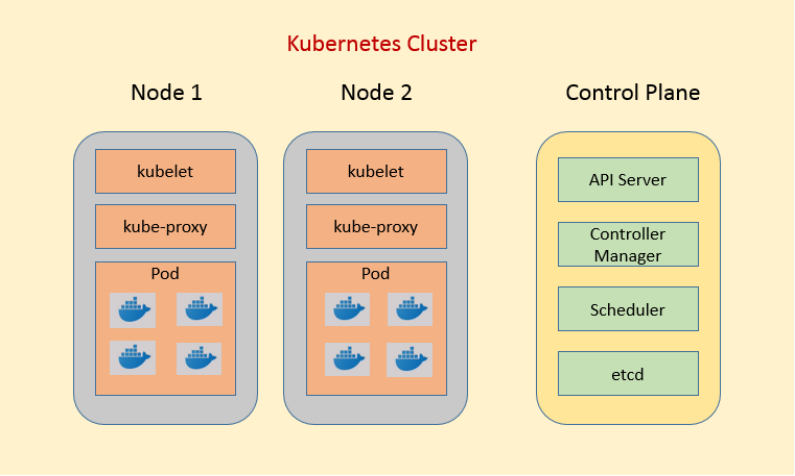
Containers

- Packages of software (application code and dependencies)
- Allow applications to be “packed” and run in any environment



Kubernetes

- Runs, monitors, and manages containers
- Schedules containers on the cluster (unit of scheduling: **pod**) and migrates them
- Load balancing across containers
- … and many more (we will see more later during the semester)



Workloads

- Memcached: distributed, in memory, key-value store - latency-sensitive
  - mcperf: load generator for memcache
- PARSEC: benchmark suite of multi-threaded applications
- iBench: suite of microbenchmarks that cause interference to various resources (CPU, caches, memory bandwidth)

Run workloads with kubectl

```
$ kubectl create -f memcache-t1-cpuset.yaml
$ kubectl expose pod some-memcached --name some-memcached-11211 \
                --type LoadBalancer --port 11211 \
                --protocol TCP
$ sleep 60
$ kubectl get service some-memcached-11211


$ kubectl get pods -o wide

NAME                READY  STATUS   RESTARTS  AGE     IP          NODE
some-memcached      1/1    Running  0         4m40s   100.96.2.3  memcache-server-z82c
```

Launch memcached

You will need this for mcperf

Yaml file for workload description

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: some-memcached
  labels:
    name: some-memcached
spec:
  containers:
    - image: anakli/memcached:t1
      name: memcached
      imagePullPolicy: Always
      command: ["/bin/sh"]
      args: ["-c", "taskset -c 0 ./memcached -t 1 -u memcache"]
  nodeSelector:
    cca-project-nodetype: "memcached"
```

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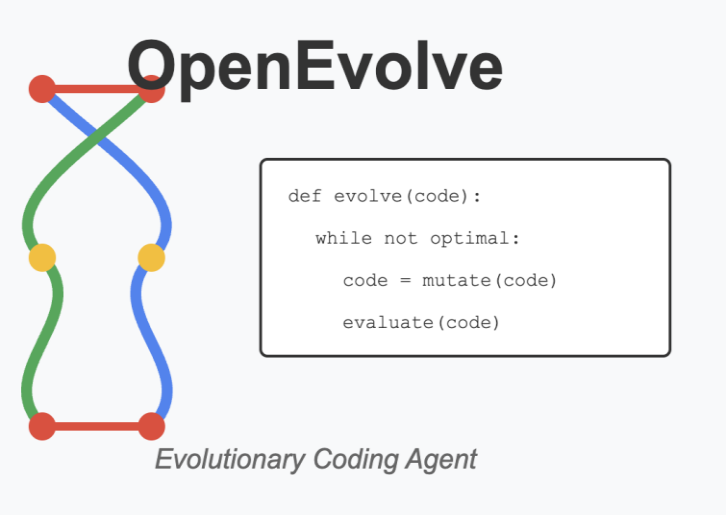
OpenEvolve Setup

OpenEvolve

- Parts 3 will include a section where you will use OpenEvolve to try to optimize the hand-crafted policies you design further with the help of generative AI
- OpenEvolve is an open source framework for automated code optimization
- OpenEvolve is based on Deepmind's AlphaEvolve
- **Rough Idea**: LLM agent will iteratively modify (evolve)
- your programs to maximize a user-defined score



OpenEvolve Setup

```
$ pip install openevolve
$ openevolve-run --config config.yaml -o <out_dir> <initial_program> <evaluator>
```

Note: We provide model access through the Swiss AI Research Platform (details in instructions)

How to get correct measurements?

Always measure one level deeper

- Thorough performance evaluation is important in:
  - **Academic research**:
    - Essential for many publications to prove the value of a new idea (prove my system is working)
  - **Industry:**
    - Maintain a high level of performance (across the lifetime) of a product
    - e.g. clients care about SLAs

- **Good performance evaluation:**
  - Deep understanding of a system's behavior - *why does the system behave like that?*
  - Understanding of internal mechanisms and components

Always Measure One Level Deeper, John Ousterhout, 2018

What is wrong with common performance evaluation practices?

What is wrong with common performance evaluation practices?

- Performance measurement is often done superficially, e.g. just before the paper deadline

- People stop measuring as soon as they find an experiment that gave positive results, omitting the less favorable results

➡ **incomplete, misleading, and erroneous measurements**

Common mistakes in performance evaluation?

Common mistakes in performance evaluation?

1. Trusting the numbers
2. Guessing instead of measuring
3. Superficial measurements
4. Confirmation Bias
5. Not enough time

Mistake 1: trusting the numbers

- Performance-measurement code is likely to have bugs, but the bugs are less obvious. These bugs could appear in:
  - The benchmarks themselves
  - The scripts that process/plot the measurements

- **Common misconception:** *If the system is not crashing, the measurements are probably correct*

- Most measurement bugs don’t cause crashes; instead they simply produce wrong numbers

Mistake 2: guessing instead of measuring

- It is unsafe to draw conclusions based on intuition without measurements to back them up
- **“What else could it possibly be?”:** not a valid justification
- It is unsafe to present an explanation as fact unless measurements confirm the specific behavior
  - For example, if you think that component A is the bottleneck, measure this and verify!
- **It is important to measure before fixing!**

Mistake 3: superficial measurements

- The **all-to-all** performance of a system is essential:
  - e.g. overall runtime, average latency of a request to the server
  - **However, this is not enough!**

- **We have to measure the "internal" behavior of a system, i.e. per-component**
  - break the performance down (e.g., what components are responsible for which portion of the latency?)
  - what improvement had the greatest impact on performance?

Mistake 4: Confirmation Bias

- Confirmation bias => select and interpret data in a way that support their hypothesis
  - you hope/expect the system will work well

- When you see a result that supports your hypothesis (e.g., “the system has high throughput”), you are more likely to accept the result without question

- In contrast, if a measurement suggests it is not performing well, you are more likely to dig deeper to explain what is happening and fix the problem

➡ **results can be inaccurate or misleading**

Mistake 5: Not enough time

- **Allow enough time to do performance evaluation!!!**
  - (almost certainly) there will be bugs
  - Many iterations of evaluation/performance improvements
  - Need to evaluate in multiple scenarios/use cases for a thorough analysis

Advice for high-quality measurements

- **Allow lots of time**
- **Don't trust the numbers:**
  - Measurements should be considered *guilty until proven innocent*
  - Take different measurements at the same level
  - Validate with back-of-the-envelope calculations/simulations
- **Use intuition to ask questions, not answer them**
  - Can help identifying directions for further exploration
  - Always validate with data before making decisions or claims
- **Always measure one level deeper**
  - Measure performance of different components
  - Look at the distribution, not only median values

Measurement infrastructure

- **Automate** measurements as much as possible
- Create a **dashboard**
  - a display to show all performance measurements from a particular run
- **Do not remove the instrumentation**
- **Presentation matters**
  - Should expose the correct level of details
  - Easy to understand
  - High-quality plotting!!

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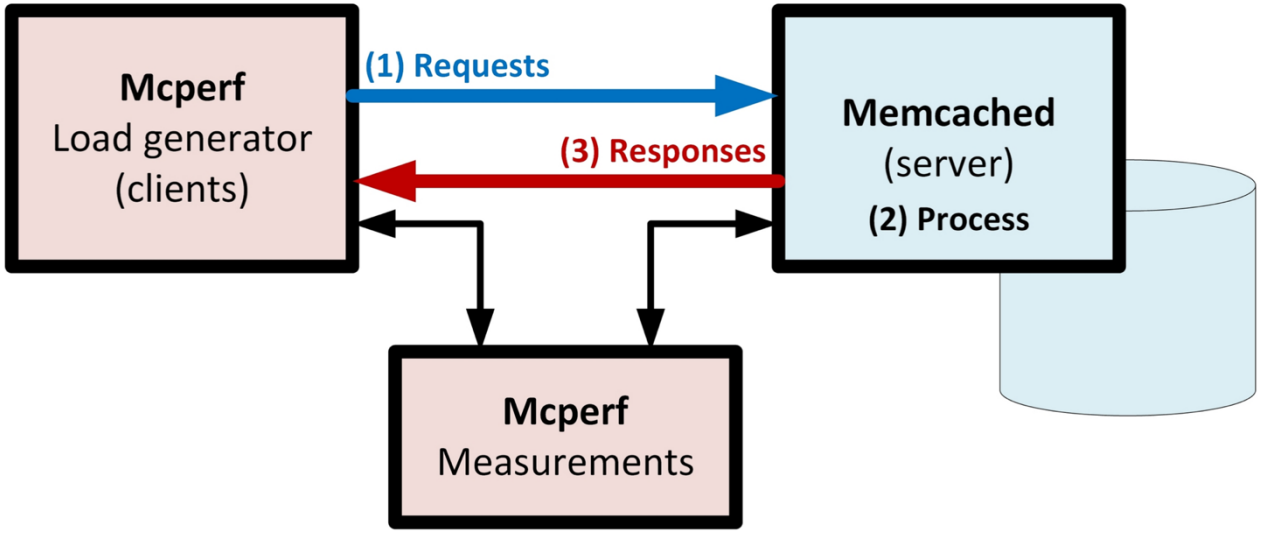
Please look at the slides in Moodle for guidelines!

Readability of your plot will be part of your grade!

ETH zürich

Plotting: common errors and guidelines

Recap: system under test (part 1)



Testing methodology (part 1)

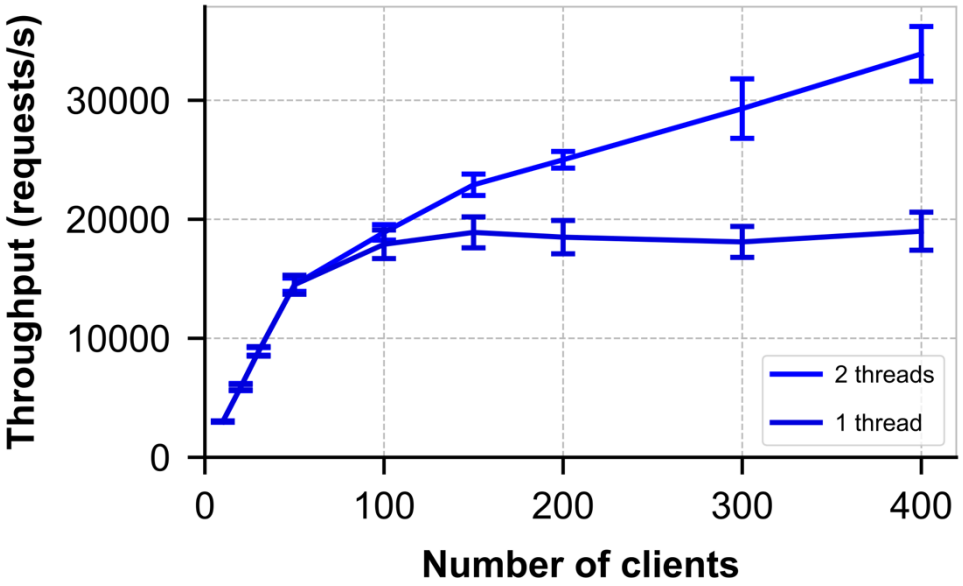
- Repeat at least 3 times
- **X-axis:** the **actual** achieved **QPS** (not the target)
- **Y-axis:** 95th percentile latency
- **Plot error bars for each point on both axes**
- Report QPS from the measurement machine

Collect and process results

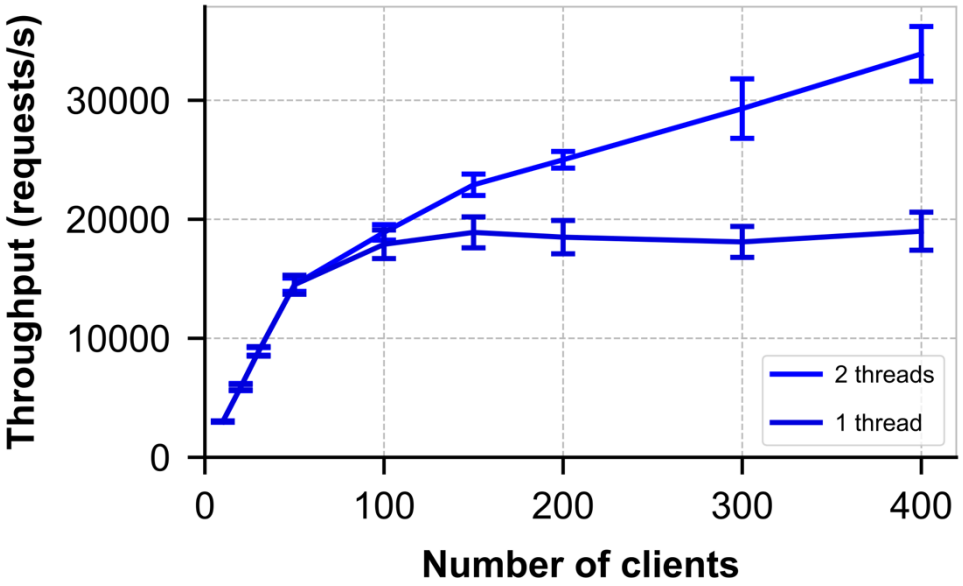
- **Structure** your measurement data and **store the results of each repetition:**
  - Collect p95 response times of each repetition
  - Collect queries per second (**achieved not target!**) of each repetition

- Compute average, sample standard deviation
- Use a scripting language that works for you: bash, Python, …
- Other statistics (e.g., min, max, percentiles, ...) might also be interesting to calculate
- **Automate** as much as possible

Plot 1: What is wrong with this plot?

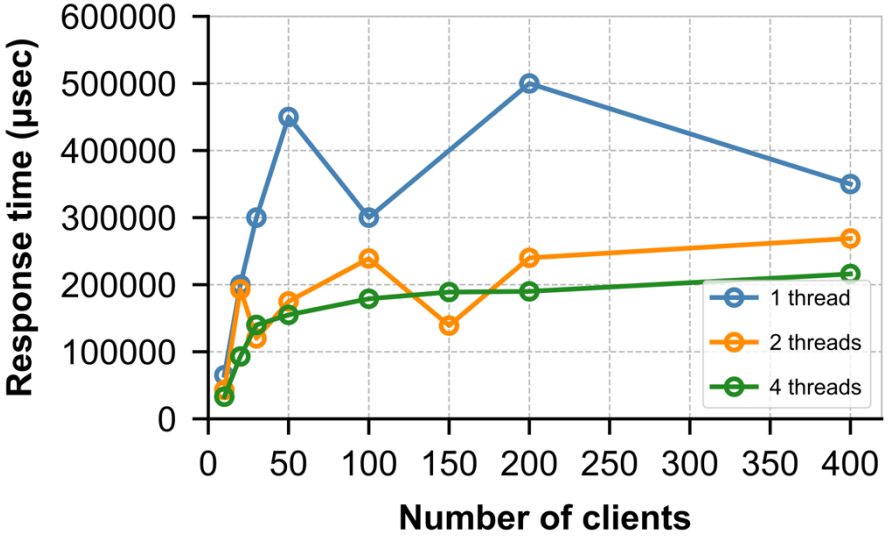


Plot 1: What is wrong with this plot?



- Misses over-saturated part
- Lines are not distinguishable (bad choice of colors)
- Missing line points

Plot 2: What is wrong with this plot?



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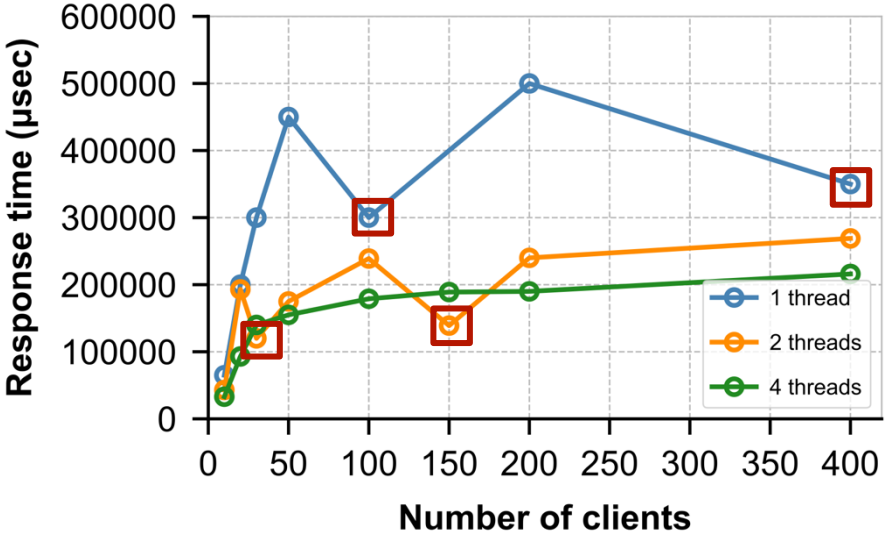

- Why these **unexpected dips** in response time?
  - Are requests being lost?
  - Is there a benchmarking problem?
  - Is the same setup used in each experiment?

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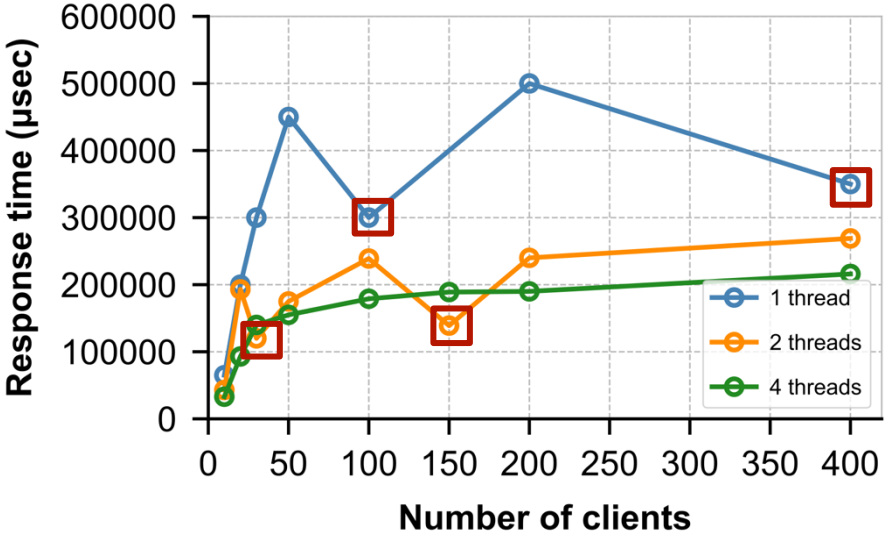
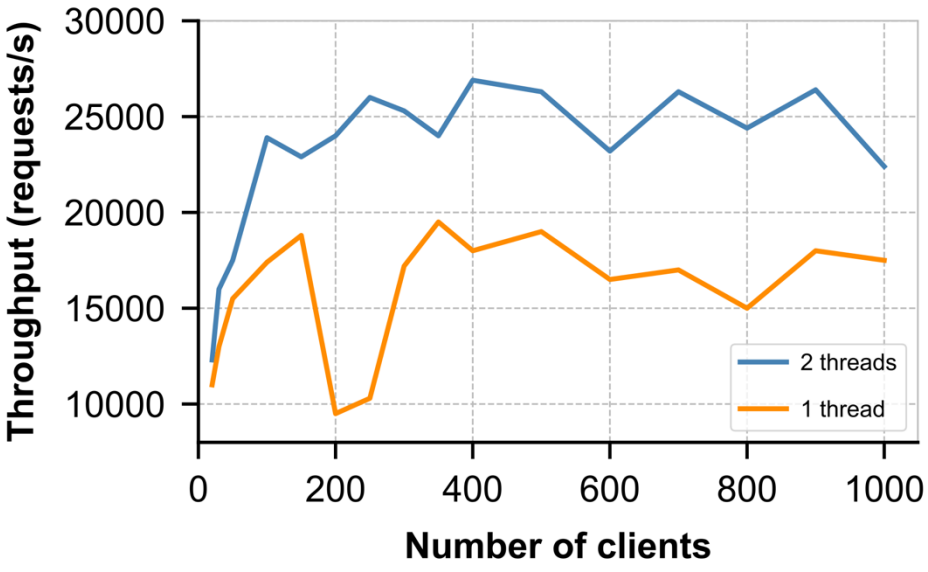


- Why these **unexpected dips** in response time?
  - Are requests being lost?
  - Is there a benchmarking problem?
  - Is the same setup used in each experiment?

- Y-axis would be better in ms-scale
- Error bars absent
- The same symbol used everywhere for the line points

Plot 3: What is wrong with this plot?



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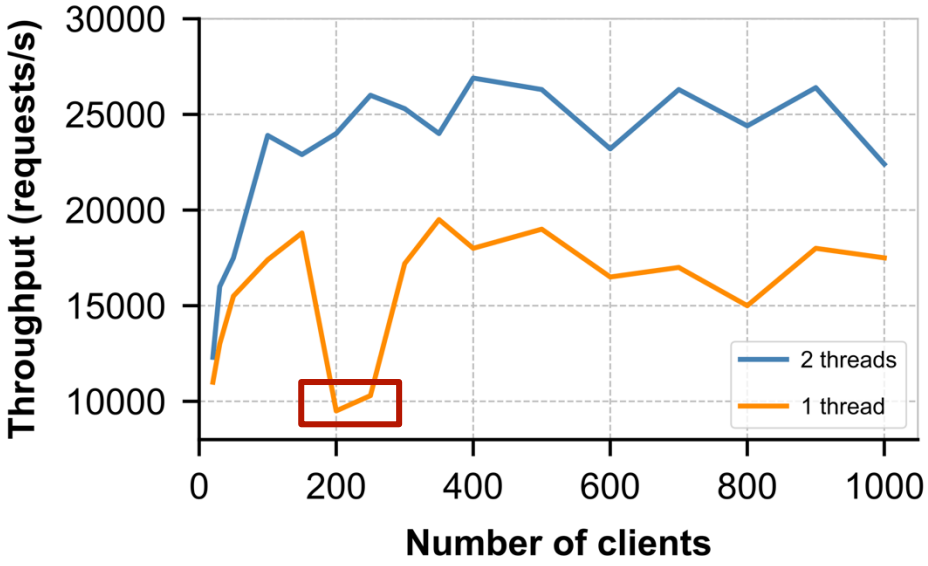
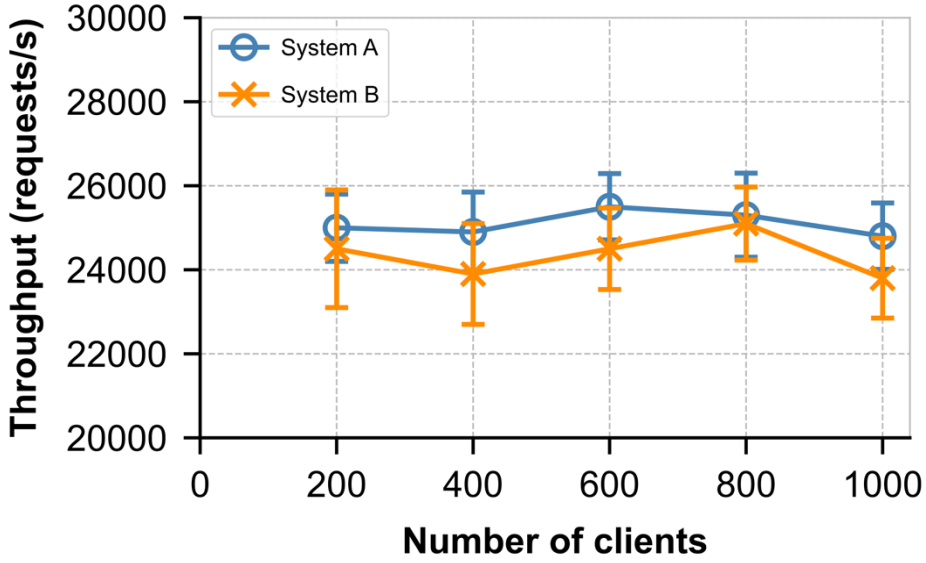


- Y-axis should start from 0
- What is going on **here**?
- Data fluctuations? Were the measurements repeated?
- Error bars and line points are absent

Plot 4: What is wrong with this plot?



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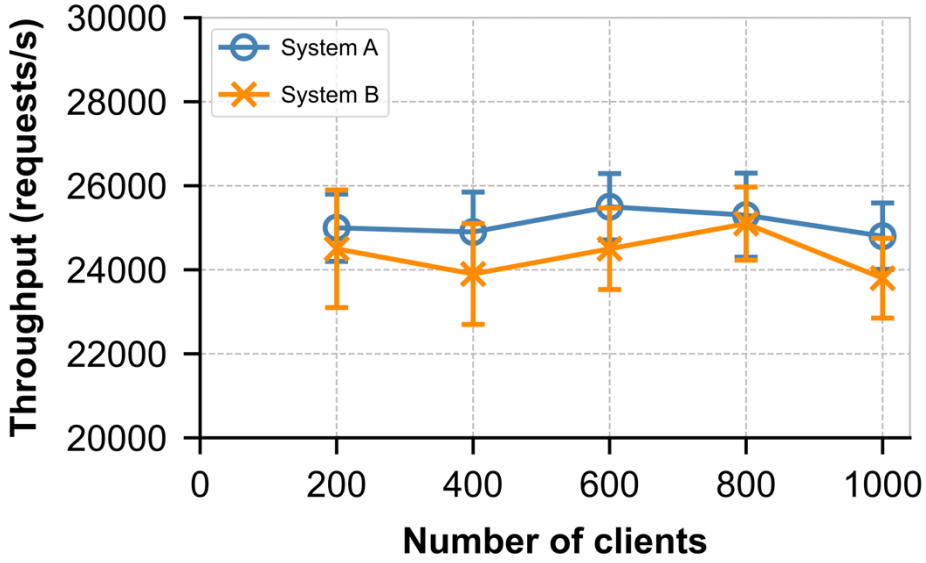
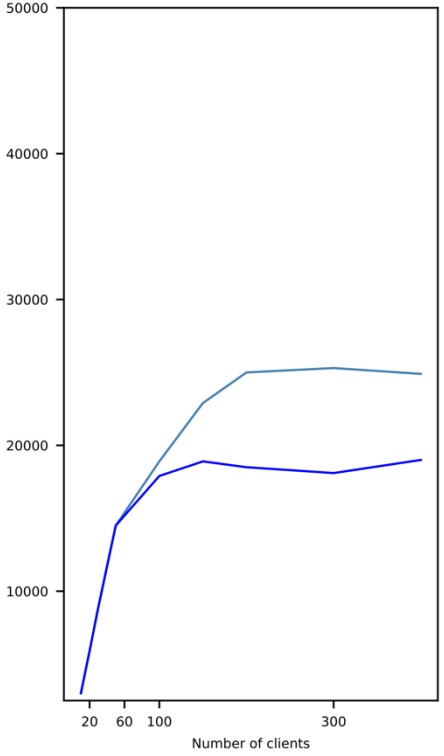


- Y-axis should start from 0
- What happens when:
  - Num_clients < 200?
  - Num_clients > 1000?
- Y-axis would be better in Krequests/s

Find the improvements



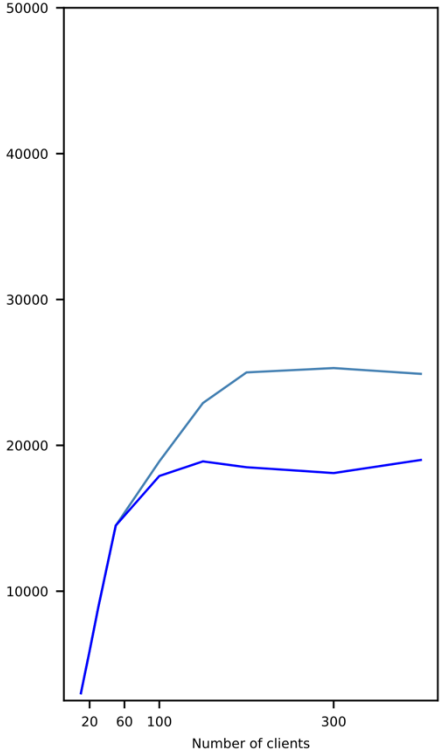
Find the improvements

Y-axis:

- Missing label
- Does not start at 0
- Upper limit too high

Inconsistent ticks at X axis

Font size too small



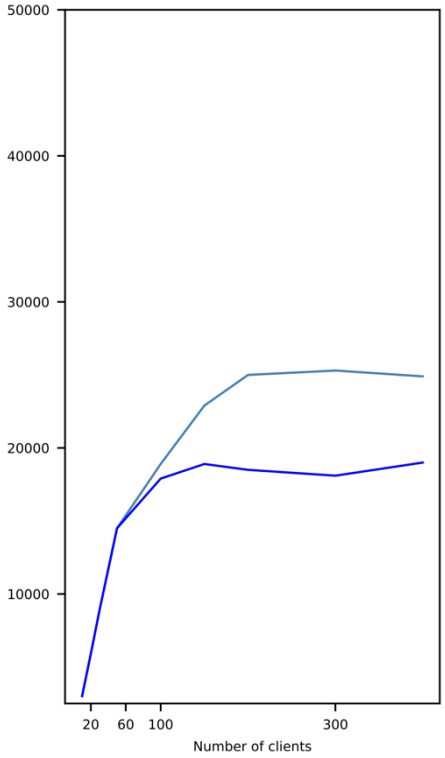
No legend

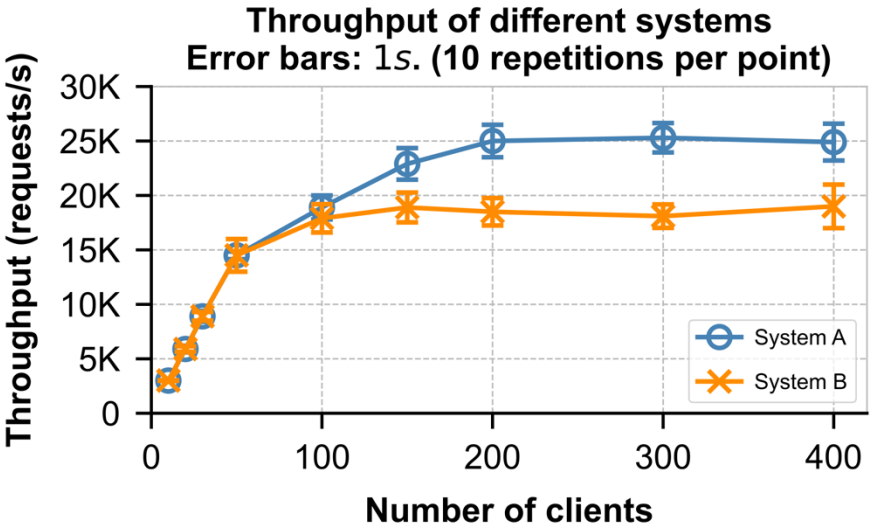
Line colors not distinguishable

Missing line points

Missing error bars

After the improvements





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What happens from a systems' perspective?

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Plotting - best practices

- **Start axis at zero**, and keep same range for related graphs
- **Label both axes, state units clearly:**
  - Use understandable units: e.g., requests/s, not requests/minute
  - Instead of 1200000 use 1.2M or 1.2 million
  - **Caution with logarithmic scales on axis**

- **Include error bars! AND**: **explain** what the error bars represent!
- Make sure system configuration is easily found: what was measured?

Benchmarking - recommendations

- As you progress and rerun experiments save results in different files:
  - you can regenerate graphs for different versions of the result

- Keep style consistent over graphs
- **Automate and use scripts for exporting graphs**

- **Plotting tools:**
  - gnuplot (shell)
  - matplotlib (Python)
  - pgfplots (LaTeX)
  - Excel (Windows) (though, not (easily) automated)

- **No hand-drawn plots!**

Example: Matplotlib

Create a plot

- import matplotlib.pyplot as plt
- fig = plt.figure()
- fig_ax = fig.gca()
- …
- plt.tight_layout()
- plt.savefig("example.pdf")

Plot data

- fig_ax.plot(x, y,…)
- fig_ax.errorbar(x,y,xerr=…,yerr=…,…)

Fix layout

- fig_ax.set_xlabel(…), fig_ax.set_ylabel(…)
- fig_ax.grid(…)
- fig_ax.set_xlim(…), fig_ax.set_ylim(…)
- fig_ax.tick_params(…)
- fig_ax.set_xticks(…), fig_ax.set_yticks(…)
- fig_ax.set_xticklabels(…), fig_ax.set_yticklabels(…)
- fig_ax.legend(…)

Thank you!
Questions?